SANKET PRAVIN DALE

Integrated B.Tech CSE Student | Software Development | Computer Vision | Hyperspectral Imaging

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PROFESSIONAL SUMMARY

Second-year Integrated B.Tech Computer Science and Engineering student at Sanjivani University with hands-on project experience in image processing, desktop application development, computer vision, and hyperspectral imaging. Interested in research and internship opportunities involving practical software solutions, spectral analysis, and technology-driven problem solving.

EDUCATION

Integrated B.Tech in Computer Science and Engineering — Sanjivani University | Currently in Second Year | Expected graduation: 2029–2031 (confirm expected completion year)

Class 10 — De Paul English Medium School, Yeola | 83% | [Year of completion]

TECHNICAL SKILLS

Programming — Python, C

Web Technologies — HTML, basic web page design

Application Development — PyQt, desktop application development

Computer Vision & Image Processing — OpenCV, image processing, Grad-CAM visualization

Machine Learning — EfficientNet-B0, foundational deep learning concepts

Hyperspectral Analysis — Region-of-interest selection, spectral data extraction, wavelength-based visualization

Hardware Integration — Intel RealSense D455f, camera-based inspection

Tools & Documentation — Excel export, technical documentation, presentation preparation

PROJECTS

Plant Disease Detection Using Hyperspectral Imaging — Worked on a plant leaf disease detection project combining image processing and deep learning. Used an EfficientNet-B0-based classification workflow and Grad-CAM visualization to inspect image regions influencing predictions. Explored wavelength-band visualization and spectral analysis across the 380–780 nm range. [Confirm exact individual contribution and whether the system uses measured hyperspectral data or simulated/derived bands from ordinary images.]

Hyperspectral Analysis System — Developed an interactive desktop application for region-based image analysis and spectral data visualization. Implemented point, rectangle, and ellipse region selection, intensity-data extraction, graph plotting, and Excel export. Worked on the PyQt interface for image loading, zooming, analysis, and results export.

HYPERLUMA – Eye Disease Screening — Contributed to a project focused on eye screening and diabetic retinopathy-related diagnostic workflows. Explored retinal imaging, clinical screening processes, and practical challenges through professional guidance and visits. Helped prepare project documentation and proposal materials. [Add exact role, dates, and verified outcomes.]

Onion Inspection and Size Classification — Worked on a camera-based onion inspection application using an Intel RealSense D455f and OpenCV. Developed image-processing functionality for onion detection and diameter/size measurement, with size categorization and a PyQt interface. Focused on camera integration and a practical inspection workflow. [Confirm the final number of size categories and your exact implementation.]

RESEARCH & TECHNICAL EXPOSURE

Hyperspectral imaging project work — Project period: 15 May–15 June 2026. Mentor/supervisor: Dr. Arvind Mukundan [confirm official title and relationship].

Ethical hacking workshop — Participated in an ethical hacking workshop at IIT Bombay. [Add date or certificate details if available.]

Cybersecurity exploration — Explored cybersecurity concepts and the Censys internet asset search platform.

LEADERSHIP & ACTIVITIES

Technical Head – Integrated B.Tech Department — Supported coordination of departmental technical activities and student initiatives. [Confirm official role title, institution/department, and dates before submitting.]

AREAS OF INTEREST

Cybersecurity and ethical hacking, software development, computer vision, hyperspectral imaging, machine learning, and research.